

PrithviSaarthi - Climate Data Analytics Platform

Project Documentation

NASA Space Apps Climate Analytics Dashboard

Overview

PrithviSaarthi is a climate data analytics platform developed to analyze global environmental trends using interactive Power BI dashboards and data visualization techniques. The project focuses on climate awareness through analytics-driven storytelling by transforming complex environmental datasets into meaningful insights.

Business Problem

Climate datasets are large, complex, and difficult to interpret without proper visualization. Organizations and researchers require interactive analytical platforms to monitor environmental changes, understand sustainability trends, and identify key climate indicators affecting global ecosystems.

Objectives

- Analyze global climate change indicators over multiple years
- Track greenhouse gas and CO₂ emission patterns
- Study global temperature rise trends
- Monitor renewable energy adoption growth
- Build interactive dashboards for climate awareness and storytelling

Tools & Technologies Used

- Power BI
- Microsoft Excel
- Data Visualization
- KPI Analytics
- Climate Data Reporting

Dataset

The project uses multiple climate-related datasets covering greenhouse gas emissions, carbon dioxide concentration levels, global temperature trends, sea level variations, renewable energy generation, and environmental indicators by country and year.

Process Workflow

- Data Collection from climate datasets
- Data Cleaning and Transformation
- KPI Identification

- Data Modeling in Power BI
- Dashboard Development
- Trend Analysis and Insight Generation

Dashboard Features

- Greenhouse Gas Emission Analysis
- CO₂ Emission Tracking
- Global Temperature Trend Visualization
- Renewable Energy Adoption Analysis
- Sea Level Trend Monitoring
- Interactive Filters and KPI Cards
- Country-wise Climate Comparison

Key KPIs

- Annual Methane Emissions
- Annual Carbon Dioxide Levels
- Global Temperature Change
- Renewable Energy Growth
- Sea Level Change Trends
- Sector-wise Greenhouse Gas Emissions

Key Insights

- Identified continuous growth in global carbon dioxide concentration over recent decades
- Observed rising global temperature trends indicating long-term climate change patterns
- Analyzed greenhouse gas emissions across different industrial sectors
- Detected increasing renewable energy adoption in recent years
- Visualized long-term sea level rise trends using historical climate records

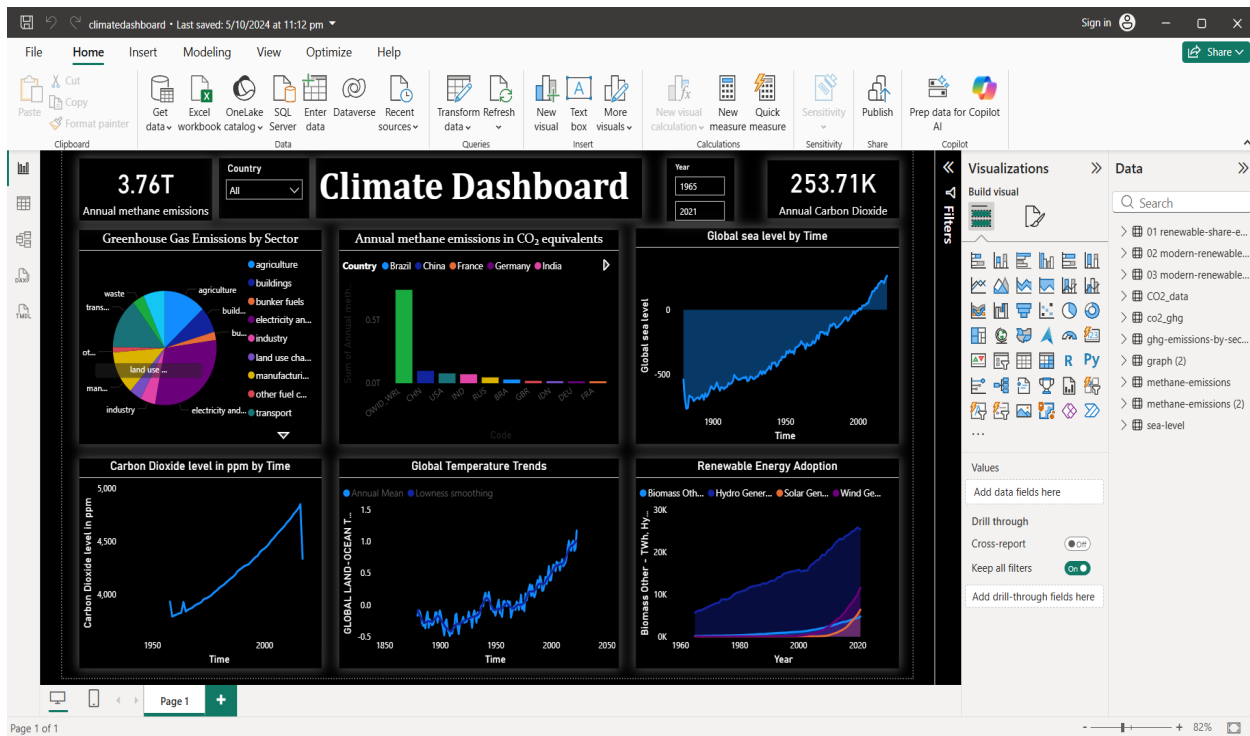
Awards & Recognition

- Galactic Impact Award – NASA Space Apps Challenge 2024
- Local Winner & Global Nominee 2025

Future Improvements

- Integration with real-time climate APIs
- Predictive climate forecasting models
- AI-driven anomaly detection
- Advanced geospatial climate analysis
- Carbon footprint recommendation system

Dashboard Preview



Conclusion

PrithviSaarathi demonstrates how data analytics and visualization can support climate awareness, sustainability discussions, and environmental decision-making through impactful storytelling and interactive dashboards.